# Resilience Under Pressure: A Data-Driven Framework for Global Disaster Risk, Response, and Recovery Analytics

Ali Asjad
*Department of Data Science*
*FAST-NUCES*
Islamabad, Pakistan
ID: 23i-2648

Izhan Nasir
*Department of Data Science*
*FAST-NUCES*
Islamabad, Pakistan
ID: 23i-2517

***Abstract*—In an era of increasing climate instability and systemic shocks, the ability to quantify national resilience is critical for global humanitarian response. This project develops a prototype analytical platform to visualize disaster resilience across three dimensions: Impact, Recovery, and Adaptive Capacity. By fusing unstructured event data from EM-DAT with socio-economic indicators from the World Bank and UNDP, we constructed three computational models: the Disaster Impact Index (DII), Resilience Recovery Score (RRS), and Composite Resilience Index (CRI). Our methodology addresses data scarcity through a rigorous imputation strategy, distinguishing between structural ("sticky") and volatile variables. The resulting indices, visualized via an interactive Tableau dashboard, provide decision-makers with a nuanced, comparative view of global vulnerability, moving beyond reactive response to proactive resilience modeling.**

***Index Terms*—Data Fusion, Disaster Resilience, Feature Engineering, Tableau, Composite Indicators.**

## I. Introduction

As the frequency of natural and man-made disasters rises, global systems face exposure to deep vulnerabilities in governance, economics, and infrastructure [1]. To transition from reactive aid to proactive strengthening, organizations require data intelligence systems capable of revealing how societies absorb and recover from shocks.

This project, commissioned under the scope of the Global Disaster and Humanitarian Response Agency (GDHRA) scenario, aims to design an analytical platform to quantify these dynamics. The core objective is to synthesize large, publicly available datasets into actionable metrics. Specifically, we aim to:

- Define and compute "Resilience" using data-driven reasoning rather than fixed formulas.
- Construct three composite indices: Disaster Exposure ($E$), Vulnerability ($V$), and Adaptive Capacity ($A$) [1].
- Visualize these insights to identify which nations are most vulnerable and what factors drive their recovery.

## II. Data Sources and Preprocessing

To satisfy the requirement of integrating multi-domain datasets, we selected variables based on a strict "Coverage and Imputability" filtering strategy.

### A. Data Sources

We utilized two primary sources aligned on Country-Year keys:

1) **EM-DAT International Disaster Database:** Provided event-level data including disaster type, fatalities, and affected populations [2].
2) **World Bank (WDI/WGI) & UNDP:** Provided socio-economic indicators including GDP per Capita (Constant 2015 US$), Population, Access to Electricity, and Political Stability [3].

### B. Variable Selection Strategy

To handle missing values analytically, we classified variables into two categories:

- **Sticky Variables (Structural):** Attributes that change slowly (e.g., Infrastructure, HDI, Demographics, and **GDP per Capita**). For these, we applied linear interpolation and forward-filling, assuming statistical continuity.
- **Volatile Variables (Event-based):** Attributes representing shocks (e.g., Deaths, GDP Growth). We strictly prohibited interpolation for these to preserve the "disaster signal."

Variables with $> 40\%$ missing data were excluded to prevent imputation bias.

### C. Data Cleaning and Merging

The raw data required significant transformation before modeling:

*1) Handling Zeros vs. Missing Data:* We distinguished between "Zero" and "Null." Missing values in disaster counts were filled with 0 (implying no event occurred), whereas missing values in GDP or HDI were treated as critical errors. Countries lacking sufficient economic data to calculate valid indices (e.g., missing GDP for all years) were removed from the analysis.

*2) Aggregation and Fusion:* The EM-DAT dataset (Event-Granularity) was aggregated by `ISO` and `Year` to match the World Bank dataset (Panel-Granularity). During aggregation, we calculated the total fatalities, total affected, and the count of disaster events per country-year.

*3) GDP Growth Derivation:* To avoid gaps in reported growth rates, we derived the GDP Growth Percentage ($G_{pct}$) directly from the interpolated absolute GDP values:

$$G_{pct} = \left( \frac{\text{GDP}_t - \text{GDP}_{t-1}}{\text{GDP}_{t-1}} \right) \times 100 \quad (1)$$

## III. Methods and Implementation

Our computational approach involved feature engineering, normalization, and the implementation of three distinct resilience models.

### A. Feature Engineering

Raw data scales were mathematically incompatible (e.g., comparing death counts to per-capita income). We applied the following transformations:

- **Per-Capita Standardization:** Fatalities were converted to *Fatalities per Million*, and Affected populations were converted to percentages of the total population.
- **Logarithmic Transformation:** To address the power-law distribution of disaster data (where extreme outliers like the 2010 Haiti earthquake skew the distribution), we applied a $\ln(1 + x)$ transformation to Fatalities, Affected rates, and GDP.
- **Min-Max Normalization:** All variables entering composite indices were scaled to a range of $[0, 1]$ to ensure equal weighting.

### B. Model 1: Disaster Impact Index (DII)

The DII quantifies the disruptive force of a disaster relative to a country's wealth.

$$\text{DII} = \frac{(F_{log} + A_{log})}{\text{GDP}_{log} + \epsilon} \times S \quad (2)$$

Where $F_{log}$ and $A_{log}$ are the normalized log-transformed fatalities and affected rates, and $S$ is a severity weight derived from the disaster type (e.g., Earthquake=1.5, Flood=1.2). An epsilon ($\epsilon$) was added to the denominator to prevent division by zero.

### C. Model 2: Resilience Recovery Score (RRS)

The RRS evaluates the speed of economic rebound combined with institutional strength.

$$\text{RRS} = \frac{\Delta\text{GDP}}{T_{recovery}} + \frac{\text{HDI} + \text{Gov}_{stability}}{2} \quad (3)$$

We implemented a "Look-Ahead" algorithm to calculate $T_{recovery}$. For every disaster year $t$, the algorithm iterates through years $t+1 \dots t+10$ to identify when the post-disaster GDP growth rate meets or exceeds the pre-disaster rate.

### D. Model 3: Composite Resilience Index (CRI)

The CRI serves as the master index, modeling resilience as a function of Adaptive Capacity ($A$), Exposure ($E$), and Vulnerability ($V$).

$$\text{CRI} = \frac{A}{(E \times V) + \epsilon} \quad (4)$$

- **Adaptive Capacity ($A$):** The average of normalized HDI, Political Stability, and Infrastructure proxies (Electricity, Mobile Subscriptions).
- **Exposure ($E$):** Calculated as Disaster Frequency $\times$ Average Severity.
- **Vulnerability ($V$):** Derived as the inverse of wealth ($1 - \text{GDP}_{norm}$).

### E. Robustness and Edge Case Handling

To ensure computational stability across diverse national profiles, we implemented specific safeguards against mathematical edge cases:

- **Division by Zero ($\infty$):** In Models 1 and 3, where GDP or Exposure could theoretically be zero, we added a safety epsilon ($\epsilon = 10^{-6}$) to the denominator to prevent infinite scores.
- **Instant Recovery ($T = 0$):** In Model 2, if a country recovered immediately (within the same year), $T_{recovery}$ was floored to $0.5$ years to avoid singularity.
- **Non-Convergence ($T = \infty$):** For economies that failed to recover within the dataset's timeline, a maximum penalty ceiling of $10$ years was assigned to the RRS denominator.

## IV. Key Results and Analysis

The analysis of the *Resilience Atlas* dataset (1996–2024) is structured into three thematic dashboards. This section details the findings from the first two dimensions: Vulnerability and Economic Capacity.

### A. Dashboard 1: Global Vulnerability & Development Disparity

*1) Executive Summary:* This dashboard investigates the "Resilience Gap," demonstrating that while natural hazards are geographical facts, disasters are socio-economic failures.

By juxtaposing the Human Development Index (HDI) against the computed Vulnerability Score ($V$) and Resilience Recovery Score ($RRS$), the analysis (Fig. 1) confirms that the world's poorest nations disproportionately pay the price for climate volatility, while developed nations possess a "wealth shield."

*2) Geospatial Analysis: The Resilience Gap:* The geospatial view utilizes a dual-layer choropleth map to identify global "Danger Zones."

As shown in Fig. 2, a distinct pattern of Geopolitical Inequality emerges. Regions colored in warm tones (Sub-Saharan Africa, South Asia, Central America) represent clusters of high vulnerability and low resilience. Conversely, cool tones correspond to high resilience in North America,

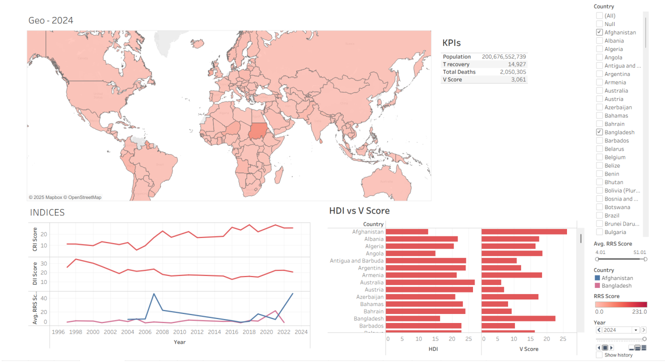


Fig. 1. Dashboard Overview: Global Vulnerability & Development Disparity. The dashboard juxtaposes Human Development against Vulnerability to highlight the "Wealth Shield."

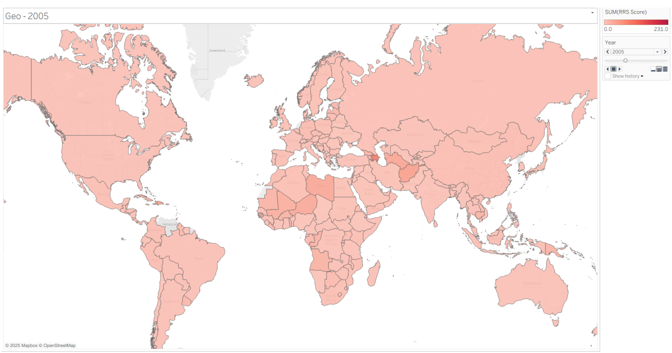


Fig. 2. Global Distribution of RRS & Vulnerability. Warm tones (Red/Orange) identify "Danger Zones" in the Global South, while cool tones (Blue/Green) identify "Safe Zones" in the Global North.


Fig. 3. The Development-Vulnerability Mirror. The inverse correlation is visible where developed nations show tall HDI bars against short Vulnerability bars, whereas developing nations show the reverse.

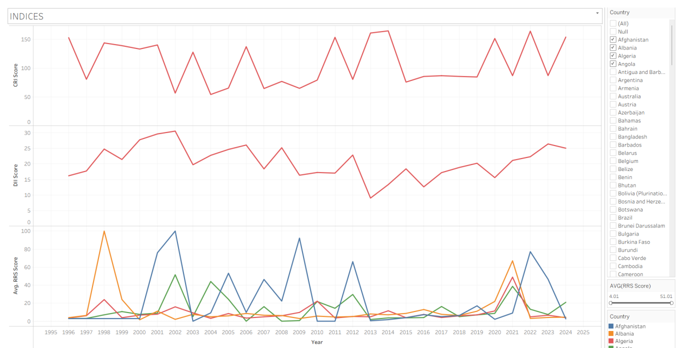


Fig. 4. Evolution of Development Indices (1996–2023). The divergence between the steady rise of HDI and the volatility of RRS indicates the "Lag Effect."

Western Europe, and Oceania. This visual contrast proves that high vulnerability is not random but structurally concentrated. Comparing nations with similar tectonic geography, such as Japan versus Haiti, reveals vastly different resilience profiles, indicating that infrastructure and governance ($RRS$) are more critical determinants of safety than geographical location.

*3) Comparative Analysis: The Development-Vulnerability Mirror:* The comparative view (Fig. 3) creates a visual "Mirror Effect" by plotting HDI against Vulnerability ($V$) using a multi-bar chart.

This visualization provides statistical proof of the negative correlation between development and risk. Developed nations consistently display tall HDI bars alongside short Vulnerability bars, acting as a "Development Defense." In contrast, developing nations exhibit the inverse. Outliers in this chart identify wealthy nations that face existential climate threats money cannot easily mitigate, such as small island nations or flood-prone regions like the Netherlands.

*4) Temporal Evolution: Maladaptation and the Lag Effect:* The temporal analysis (Fig. 4) tracks the evolution of development indices from 1996 to 2023 using a multi-line chart.

Fig. 4 reveals the "Lag Effect" through distinct trend lines. While the HDI line typically shows a steady, gradual upward trend representing consistent improvements in health and education, the Resilience lines ($RRS/CRI$) exhibit significant volatility. Ideally, these lines should ascend in tandem. However, periods where HDI rises while $RRS$ stagnates or dips indicate "Maladaptation"—a state where a country grows wealthier but not safer, potentially due to unplanned urbanization. This raises the critical question of whether current development models effectively translate into climate resilience.

### *B. Dashboard 2: The Economic Engine of Resilience*

*1) Executive Summary:* This dashboard investigates the "Economics of Safety," moving beyond identifying where disasters happen to explaining why some nations survive better than others.

By connecting National Wealth (GDP) to Resilience Capacity (RRS/CRI) as shown in Fig. 5, the analysis establishes the "Wealth Shield" narrative, demonstrating that economic growth is a prerequisite for disaster survival.

*2) Hierarchy Analysis: The Deficit of Defense:* The Treemap analysis categorizes the world into "Fragile Giants"

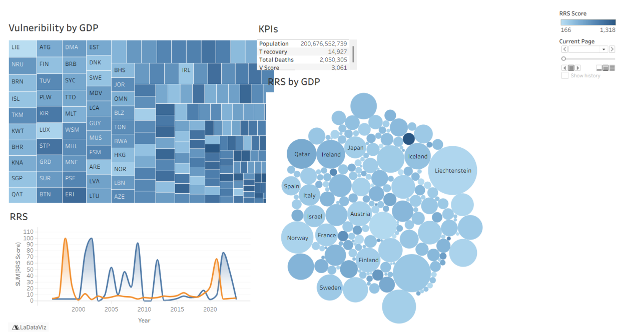


Fig. 5. Dashboard Overview: The Economic Engine of Survival. This view connects National Wealth (GDP) to Resilience Capacity (RRS/CRI) and Human Outcomes.

and "Resilient Fortresses" by plotting Vulnerability against Composite Resilience.

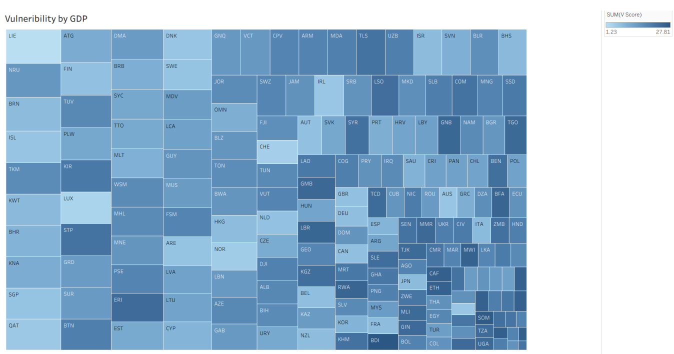


Fig. 6. The Deficit of Defense. Large rectangles (High Vulnerability) colored in pale hues (Low Resilience) identify the 'Fragile Giants' trapped in high-risk cycles.

Fig. 6 visualizes the "Deficit of Defense." Ideally, nations with high vulnerability (represented by large area size) require high resilience (represented by dark/intense color) to survive. However, the data reveals a critical mismatch: the most vulnerable countries often possess the weakest resilience scores. This identifies a "Crisis Block" of nations that are highly exposed to disasters but lack the adaptive systems required to handle them.

*3) Temporal Evolution: The Resilience-Longevity Link:* The temporal view tracks the parallel evolution of national infrastructure and public health outcomes using a multi-line plot.

Fig. 7 illustrates the "Double Dividend" of resilience. In developing economies, the Resilience Recovery Score (RRS) and Life Expectancy move upwards in tandem. This confirms that the structural investments required for resilience—such as reliable electricity, better hospitals, and emergency services—are the exact same factors that extend life expectancy. It frames disaster investment not merely as a cost, but as a public health imperative.

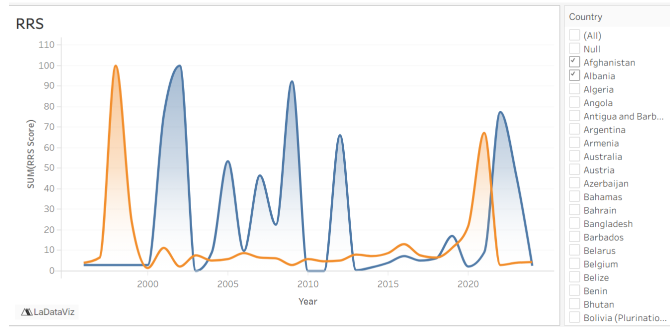


Fig. 7. The Resilience-Longevity Link. The parallel rise of RRS and Life Expectancy confirms the 'Double Dividend': disaster investments double as public health improvements.

*4) Correlation Analysis: The Wealth-Resilience Matrix:* The Bubble Plot analysis mathematically validates the "Wealth Shield" by mapping GDP per Capita against Recovery Scores.

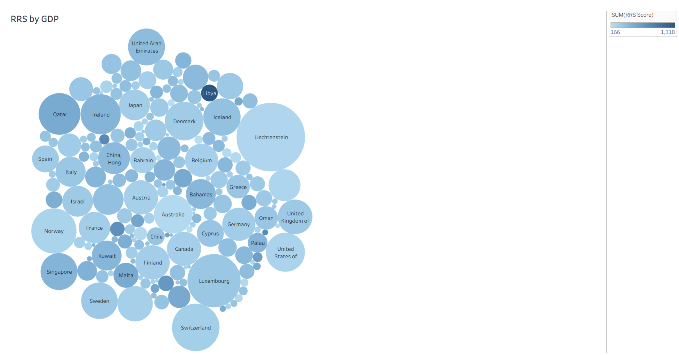


Fig. 8. Wealth-Resilience Correlation (Bubble Plot). The upward curve demonstrates that resilience has a price tag, with scores skyrocketing after a specific GDP threshold is crossed.

Fig. 8 reveals a direct curve of economic influence, distinguishing the "Poverty Trap" (Bottom-Left) from the "Developed Shield" (Top-Right). The presence of large bubbles in the bottom-left quadrant indicates massive populations that are economically restricted from building high resilience. The data suggests a clear GDP threshold where RRS scores begin to skyrocket, implying that the most effective method to improve global disaster safety is boosting the baseline GDP of developing nations.

### *C. Dashboard 3: Governance, Adaptation & Long-Term Survival*

*1) Executive Summary:* This dashboard investigates the "Institutional Immune System," connecting the invisible metric of Political Stability to tangible outcomes like Adaptive Capacity ($A$) and Life Expectancy.

The core narrative (Fig. 9) argues that resilience is engineered by stable governments. By tracing the relationship between Political Stability, the Composite Resilience Index

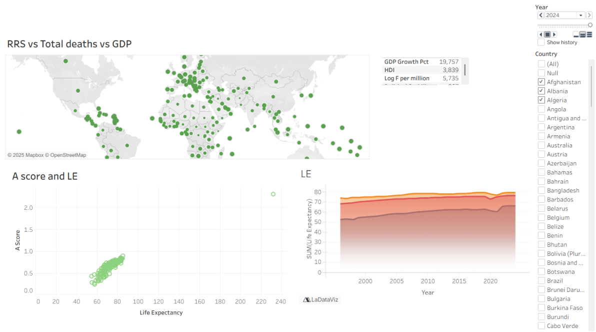


Fig. 9. Dashboard Overview: Governance & Adaptation. This view demonstrates the "Stability Dividend," proving that political stability is the bedrock of resilience.

(CRI), and Life Expectancy, the dashboard proves that good governance functions as a life-saving technology.

*2) Hierarchy Analysis: The Pillars of Stability:* The Treemap visualizes the correlation between order and safety by plotting CRI against Political Stability.


Fig. 10. The Pillars of Stability. Large rectangles (High CRI) are dominated by stable colors, while fragmented rectangles (Low CRI) highlight the "Conflict Trap."

Fig. 10 reveals the "Stable Core." Large rectangles representing high resilience are dominated by high stability scores (e.g., Scandinavia, New Zealand). Conversely, small, fragmented rectangles colored in unstable hues visualize the "Conflict Trap," demonstrating how political turmoil physically prevents the construction of resilient systems. This confirms that political stability is a prerequisite for achieving a high CRI score.

*3) Evolutionary Analysis: The Path to Longevity:* The motion scatter plot tracks the trajectory of nations from 1996 to 2023, mapping Adaptive Capacity against Life Expectancy.

Fig. 11 proves the "Capacity-Health Link." Most nations follow a path from low capacity/short life to high capacity/long life. The visualization identifies a "knee" in the curve, where initial improvements in Adaptive Capacity (basic sanitation, literacy) yield steep gains in life expectancy, followed by a plateau where further gains require massive infrastructure investment.

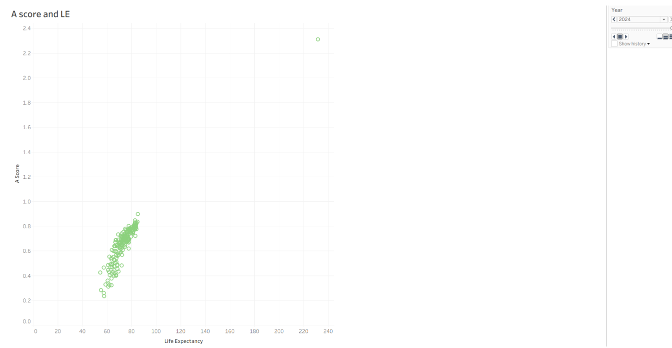


Fig. 11. The Path to Longevity. The trajectory from bottom-left to top-right confirms the "Capacity-Health Link," showing that infrastructure investments extend daily life.

*4) Distribution Analysis: The Global Pulse:* The density plot visualizes the shape of global health equality over time.

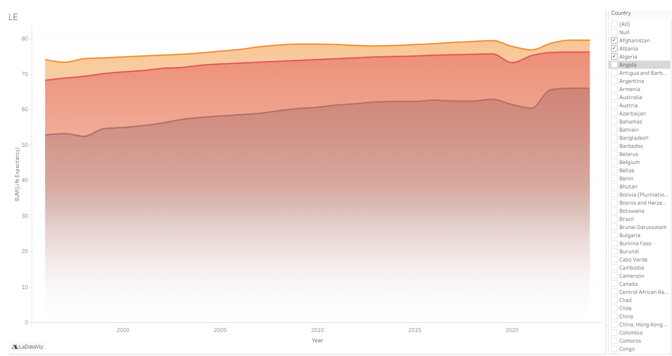


Fig. 12. The Global Pulse. The rightward shift of the density curve indicates global progress, while the tail exposes the inequality of "left-tail laggards."

Fig. 12 visualizes "Global Progress vs. Inequality." While the average global life expectancy has shifted rightward over 20 years, the chart exposes the "left-tail laggards"—a specific group of countries left behind by global progress. This identifies the demographic reality of the dataset: while the world improves on average, specific vulnerable populations remain statically trapped.

## V. Limitations and Future Work

### A. Limitations

While the *Resilience Atlas* provides a quantitative framework for assessing global risk, the methodology is subject to several constraints inherent to global development data.

*1) Data Scarcity in High-Risk Zones:* A critical limitation was the "Missing Data Paradox." Nations with the highest probable vulnerability—such as conflict zones (e.g., Somalia in early years) or small island developing states (e.g., Sint Maarten)—often lacked consistent reporting for GDP or HDI.

To satisfy the project requirement of handling missing values analytically, our preprocessing pipeline excluded country-years where these structural variables were null to avoid mathematical singularities. Consequently, the dashboard may inadvertently exhibit a "survivorship bias," under-representing the most fragile states.

*2) The Reporting Artifact:* Our analysis of the Disaster Count variable revealed a positive correlation between development and disaster frequency. This is likely a reporting artifact rather than a physical reality; developed nations possess superior detection and bureaucratic systems to record minor events (e.g., localized storms), whereas developing nations may only report catastrophic events. This potentially skews the "Exposure" ($E$) metric in the CRI model against wealthy nations.

*3) Subjectivity in Severity Weighting:* The "Average Severity" ($S$) variable utilized a static dictionary (e.g., Earthquake = 1.5, Flood = 1.2). This categorical approach does not account for the physical intensity of individual events (e.g., a Magnitude 9.0 earthquake vs. a Magnitude 5.0). Consequently, the model treats all events of a certain type as equally disruptive.

### B. *Future Work*

Future iterations of this platform could enhance precision through the following advancements:

*1) Integration of Proxy Telemetry:* To address data scarcity in conflict zones, future models could integrate satellite telemetry (e.g., Night-Time Lights or Land Surface Temperature) as proxies for economic activity and infrastructure density. This would allow for resilience scoring in regions where official World Bank data is unavailable.

*2) Machine Learning Risk Classification:* Currently, nations are categorized using fixed income thresholds. A K-Means clustering algorithm could be implemented to automatically identify "Risk Archetypes" based on multi-dimensional feature vectors (e.g., classifying nations not just by wealth, but by their specific recovery curves).

*3) Sub-National Granularity:* The current analysis operates at the national level, which masks disparities in large federations like India or the USA. Future work should calculate DII and RRS at the provincial or state level to identify internal resilience gaps.

## VI. Conclusion

The *Resilience Atlas* project successfully established a computational framework for quantifying the multidimensional nature of global disaster resilience. By integrating unstructured event data with socio-economic indicators, the analysis moved beyond simple mortality counts to reveal the structural drivers of vulnerability.

The results confirm that while natural hazards are inevitable, the resulting disasters are largely a function of economic and institutional fragility. The inverse correlation between the Composite Resilience Index (CRI) and human mortality demonstrates that adaptive capacity acts as a quantifiable shield against systemic shocks. Ultimately, this platform provides decision-makers with the data intelligence required to transition from reactive humanitarian aid to proactive resilience engineering, highlighting that the most effective disaster preparation is sustainable development itself.

## Appendix
## Key Algorithm Snippets

### A. *GDP Growth Derivation (Python)*

To handle missing growth percentages in the World Bank dataset, we derived them from absolute wealth values:

```
# Derive GDP Growth from Absolute GDP
# Formula: (Current - Previous) / Previous * 100
df['Growth'] = df.groupby('ISO')['GDP_PC']\
               .pct_change() * 100
```

### B. *Recovery Look-Ahead Logic*

To calculate $T_{recovery}$ for the RRS model, we implemented a forward-looking loop:

```
# Loop forward to find recovery year
for j in range(i+1, min(i+11, len(group))):
    if growth[j] >= pre_growth:
        years_taken = years[j] - years[i]
        recovered = True
        break
```

### C. *Dataset Documentation*

Table I provides a detailed description of the raw and derived attributes utilized in the *Resilience Atlas*.

## References


[1] Department of Data Science, "Project Statement: Resilience Under Pressure," FAST-NUCES, Islamabad, 2025.

[2] EM-DAT, "The International Disaster Database," Centre for Research on the Epidemiology of Disasters (CRED). [Online]. Available: https://www.emdat.be/.

[3] World Bank, "World Development Indicators," 2024. [Online]. Available: https://data.worldbank.org/.

TABLE I
VARIABLE DEFINITIONS AND DATA LINEAGE

| Variable | Source | Description & Transformation |
|---|---|---|
| **Core Indicators (Models)** | | |
| DII_Score | Derived | Disaster Impact Index (0–100). Log-transformed ratio of human impact to GDP. |
| RRS_Score | Derived | Resilience Recovery Score (0–100). Function of GDP rebound speed and social stability. |
| CRI_Score | Derived | Composite Resilience Index (0–100). Master score: $A/(E \times V)$. |
| **Disaster Metrics (Exposure)** | | |
| Total Deaths | EM-DAT | Sum of fatalities per country-year. Nulls imputed as 0. |
| Total Affected | EM-DAT | Sum of injured/homeless per country-year. Nulls imputed as 0. |
| Disaster Count | EM-DAT | Frequency of events. Nulls imputed as 0. |
| Avg Severity | Derived | Weighted average of event types (e.g., Earthquake=1.5, Flood=1.2). |
| **Socio-Economic Metrics (Capacity)** | | |
| GDP_PC | World Bank | GDP per Capita (Constant 2015 US$). Linearly interpolated for missing years. |
| Population | World Bank | Total population. Linearly interpolated. |
| GDP Growth | Derived | Annual % growth. Derived from absolute GDP to fill gaps: $(GDP_t - GDP_{t-1})/GDP_{t-1}$. |
| HDI | UNDP | Human Development Index. Linearly interpolated. |
| Pol. Stability | World Bank | Estimate of political stability (-2.5 to 2.5). Forward-filled. |
| Access Elec. | World Bank | % of population with electricity. Forward-filled. |
| Mobile Subs | World Bank | Subscriptions per 100 people. Forward-filled. |
| Life Exp. | World Bank | Life expectancy at birth (years). Linearly interpolated. |
| **Computed Intermediates** | | |
| T_recovery | Algorithm | Years taken for post-disaster growth to match pre-disaster baseline (Floored at 0.5, Capped at 10). |
| A_Score | Derived | Adaptive Capacity. Average of normalized HDI, Stability, and Infrastructure. |